🖥️ 📊 FINAL ADMIN PANEL (FULL CODE)

📁 File name:

```text
admin_panel.py
```

✅ COMPLETE CODE

```python
import tkinter as tk
from tkinter import messagebox
```

```
import mysql.connector


# ================= DATABASE CONNECTION =================

def connect_db():
    try:
        return mysql.connector.connect(
            host="localhost",
            user="root",
            password="YOUR_PASSWORD",   # 🔴 CHANGE THIS
            database="library_db"       # 🔴 MUST MATCH
        )
    except Exception as e:
        messagebox.showerror("DB Error", str(e))
        return None
```

```
# ================= ADD USER =================
def add_user():
    user_id = entry_user_id.get().strip()
    name = entry_user_name.get().strip()

    if not user_id or not name:
        messagebox.showwarning("Input Error", "All fields required")
        return

    db = connect_db()
    if not db:
        return

    cursor = db.cursor()
```

```
    try:
        # Check duplicate
        cursor.execute("SELECT * FROM users
WHERE id=%s", (user_id,))
        if cursor.fetchone():
            messagebox.showerror("Error", "User
already exists")
            return

        # Insert
        cursor.execute(
            "INSERT INTO users (id, name) VALUES
(%s, %s)",
            (user_id, name)
        )
        db.commit()
```

```
        messagebox.showinfo("Success", "User Added Successfully")

        entry_user_id.delete(0, tk.END)
        entry_user_name.delete(0, tk.END)

    except Exception as e:
        messagebox.showerror("Error", str(e))

    finally:
        db.close()


# ================== ADD BOOK ==================
def add_book():
    book_id = entry_book_id.get().strip()
```

```
    name = entry_book_name.get().strip()


    if not book_id or not name:
        messagebox.showwarning("Input Error", "All
fields required")
        return


    db = connect_db()
    if not db:
        return


    cursor = db.cursor()


    try:
        # Check duplicate
        cursor.execute("SELECT * FROM books
WHERE id=%s", (book_id,))
```

```
        if cursor.fetchone():
            messagebox.showerror("Error", "Book
already exists")
            return

        # Insert
        cursor.execute(
            "INSERT INTO books (id, name, available)
VALUES (%s, %s, %s)",
            (book_id, name, 1)
        )
        db.commit()

        messagebox.showinfo("Success", "Book Added
Successfully")

        entry_book_id.delete(0, tk.END)
        entry_book_name.delete(0, tk.END)
```

```
    except Exception as e:
        messagebox.showerror("Error", str(e))

    finally:
        db.close()



# ================== VIEW RECORDS
==================
def view_records():
    db = connect_db()
    if not db:
        return

    cursor = db.cursor()
```

```
    try:
        cursor.execute("SELECT * FROM users")
        users = cursor.fetchall()

        cursor.execute("SELECT * FROM books")
        books = cursor.fetchall()

        text_output.delete(1.0, tk.END)

        text_output.insert(tk.END, "=== USERS ===\n")
        for u in users:
            text_output.insert(tk.END, f"{u[0]} - {u[1]}\n")

        text_output.insert(tk.END, "\n=== BOOKS
===\n")
        for b in books:
            text_output.insert(
```

```
            tk.END,
            f"{b[0]} - {b[1]} (Available: {b[2]})\n"
        )

    except Exception as e:
        messagebox.showerror("Error", str(e))

    finally:
        db.close()



# ================== DELETE USER ==================
def delete_user():
    user_id = entry_user_id.get().strip()

    if not user_id:
```

```
        messagebox.showwarning("Input Error", "Enter
User ID")
        return

    db = connect_db()
    if not db:
        return

    cursor = db.cursor()

    try:
        cursor.execute("DELETE FROM users WHERE
id=%s", (user_id,))
        db.commit()

        messagebox.showinfo("Success", "User
Deleted")
```

```
    except Exception as e:
        messagebox.showerror("Error", str(e))

    finally:
        db.close()



# ================== DELETE BOOK ==================
def delete_book():
    book_id = entry_book_id.get().strip()

    if not book_id:
        messagebox.showwarning("Input Error", "Enter Book ID")
        return
```

```
    db = connect_db()
    if not db:
        return

    cursor = db.cursor()

    try:
        cursor.execute("DELETE FROM books
WHERE id=%s", (book_id,))
        db.commit()

        messagebox.showinfo("Success", "Book
Deleted")

    except Exception as e:
        messagebox.showerror("Error", str(e))
```

```
    finally:
        db.close()



# ================= GUI =================
root = tk.Tk()
root.title("Library Admin Panel")
root.geometry("500x600")

# ===== USER SECTION =====
tk.Label(root, text="Add User", font=("Arial", 14,
"bold")).pack()

entry_user_id = tk.Entry(root)
entry_user_id.pack()
entry_user_id.insert(0, "User ID (U001)")
```

```
entry_user_name = tk.Entry(root)
entry_user_name.pack()
entry_user_name.insert(0, "User Name")

tk.Button(root, text="Add User",
command=add_user).pack(pady=5)
tk.Button(root, text="Delete User",
command=delete_user).pack(pady=5)

# ===== BOOK SECTION =====
tk.Label(root, text="Add Book", font=("Arial", 14,
"bold")).pack()

entry_book_id = tk.Entry(root)
entry_book_id.pack()
entry_book_id.insert(0, "Book ID (B001)")

entry_book_name = tk.Entry(root)
```

```
entry_book_name.pack()

entry_book_name.insert(0, "Book Name")


tk.Button(root, text="Add Book",
command=add_book).pack(pady=5)

tk.Button(root, text="Delete Book",
command=delete_book).pack(pady=5)


# ===== VIEW =====

tk.Button(root, text="View Records",
command=view_records).pack(pady=10)


# ===== OUTPUT BOX =====

text_output = tk.Text(root, height=15)

text_output.pack()


root.mainloop()
```

🔴 🔧 WHAT YOU MUST CHANGE (VERY IMPORTANT)

✅ 1. Database Password

```python
password="YOUR_PASSWORD"
```

👉 Replace with your MySQL password

✅ 2. Database Name

```python
database="library_db"
```

👉 Must match your DB

✅ 3. Table Names MUST MATCH

```sql
users
books
```

▶️ HOW TO RUN

```bash
python3 admin_panel.py
```

🧠 HOW THIS CONNECTS TO MAIN SYSTEM

🔹 Admin Panel inserts data:

```sql
INSERT INTO users VALUES ('U001','Atharv');
```

🔹 Main system reads:

```python
SELECT * FROM users WHERE id='U001'
```

👉 That’s the connection 🔥

🧪 TEST IT

Add User:

```
U001
Atharv
```

Add Book:

```
B001
Python Book
```

```

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### Click:

```
View Records
```

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```